Lab 02 –

Python Operators And Basic Programming Constructs

Programming Exercise:

1.Write a program that asks the user for his name and then welcomes him. The output should look like this:

Enter your name: FE-student

Hello FE-Student

Source Code:

```
#ASK THE USER FOR THEIR NAME
name=input("Enter your name:")
#PRINT A WELCOME MESSAGE
print("Hello",name)
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\Users\hp\PycharmProjects\main\First Program.py"
Enter your name:FE-Student
Hello FE-Student

Process finished with exit code 0
```

2.Write a program which accept principle, rate and time from use and print the simple interest. The formula to calculate the simple interest is:

Simple interest=principle*rate*time/100

Source Code:

```python
principle=float(input("Enter the principle amount:"))
rate=float(input("Enter the rate of interest(in %):"))
time=float(input("Enter the time (in years):"))
#CALCULATE SIMPLE INTEREST
simple_interest=(principle*rate*time)/100
print("Simple Interest:",simple_interest)
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\Users\hp\PycharmProjects\main\First Program.py"
Enter the principle amount:1000
Enter the rate of interest(in %):5
Enter the time (in years):2
Simple Interest: 100.0

Process finished with exit code 0
```

3.Write a program that prompts the user to input number of calls and calculate the monthly telephone bills as per the following rule:

Minimum Rs. 2000 for up to 100 calls.

Plus, Rs. 0.60 per call for next 50 calls.

Plus, Rs. 0.50 per call for next 50 calls.

Plus, Rs. 0.40 per call for any call beyond 200.

Source Code:

```python
num_calls=int(input("Enter the number of calls:"))
#Initialize the total bill
bill=0
#Calculate the bill based on the number of calls
if num_calls <= 100:
       bill=2000
elif num_calls <= 150:
        bill= 2000 + (num_calls - 100) * 0.60
elif num_calls <= 200:
        bill= 2000 + (50 * 0.60) + (num_calls - 150) * 0.50
else:
        bill= 2000 + (50 * 0.60) + (50 * 0.50) + (num_calls - 200) * 0.40

#Print the total bill
print("Monthly telephone bill is: Rs.", bill)
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\Users\hp\PycharmProjects\main\First Program.py"
Enter the number of calls:200
Monthly telephone bill is: Rs. 2055.0

Process finished with exit code 0
```

4.Write a program that prompts the user to enter number in two variables and swap the contents of the variables.

Source Code:

```python
#PROMPT THE USER TO ENTER TWO NUMBERS
num1=float(input("Enter the first number:"))
num2=float(input("Enter the second number:"))
#SWAP THE CONTENTS OF THE VARIABLE
num1,num2=num2,num1
#PRINT THE SWAPPED VALUES
print("After swapping:")
print("First number:",num1)
print("Second number",num2)
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\Users\hp\PycharmProjects\main\First Program.py"
Enter the first number:5
Enter the second number:10
After swapping:
First number: 10.0
Second number 5.0

Process finished with exit code 0
```

- Write python algebraic expressions corresponding to the following statements:

(a) The sum of first five positive integers.

```python
#The sum of first five positive integers
sum_first_five=1+2+3+4+5
print(sum_first_five)
```

(b) The average age of Sara (age 23), Mark (age19), and Fatima (age 31)

```python
average_age = (23 + 19 + 31) / 3
```

(c) The number of times 73 goes into 403

```python
times_divided = 403 // 73
```

(d) The remainder when 403 is divided by 73

```python
remainder = 403 % 73
```

(e) 2 to the 10$^{\text{th}}$ power

```python
power_of_two = 2 ** 10
```

(f) The absolute value of the difference between Sara’s height (54 inches) and Marks’s height (57 inches)

```python
height_difference = abs(54 - 57)
```

(g) The lowest price among the following prices: $34.99, $29.95, and $31.50

```python
lowest_price = min(34.99, 29.95, 31.50)
```